

Shh: Secretly Managing Your SSH Keys

Shh is an elegant command-line tool designed for **securely managing SSH keys and secrets** with **AWS Secrets Manager**. It ensures **seamless, automated, and encrypted** storage and retrieval of sensitive credentials, making your DevOps workflow more secure and efficient.

Features

-  **Secure SSH Key Storage** – Store and retrieve SSH private keys securely from AWS Secrets Manager.
-  **Fast & Efficient** – Handles key injection into **ssh-agent** on the fly without writing to disk.
-  **Seamless Integration** – Works effortlessly with AWS, Ansible, and Terraform.
-  **Advanced Metadata** – Tracks key details, versions, and automatic rotation schedules.
-  **Key Rotation** – Monitors key age and suggests rotation timeframes for enhanced security.
-  **Public Key Support** – Upload **.pub** keys alongside private keys for seamless key management.
-  **Region Flexibility** – Configure AWS regions via CLI arguments or environment variables.

Installation

Prerequisites

- AWS CLI installed and configured with appropriate permissions
- **jq** for JSON processing
- **ssh-agent** running on your system

Basic Installation

Clone the repository and place the scripts in your system's **PATH**:

```
git clone https://github.com/your-username/shh.git
cd ssh
chmod +x ssh ssh-add ssh-admin
sudo mv ssh /usr/local/bin/
sudo mv ssh-add /usr/local/bin/
sudo mv ssh-admin /usr/local/bin/
```

Environment Configuration

You can configure Shh with environment variables:

```
# Regional configuration (in order of precedence)
export SHH_REGION="us-east-2" # Preferred region
export AWS_REGION="us-east-2" # Alternative
export AWS_DEFAULT_REGION="us-east-2" # AWS CLI default
```

```
# Secret name configuration
export SSH_SECRETS="my-ssh-keys" # Name of your AWS Secrets Manager
secret
```

Usage

Securely Add an SSH Key to AWS Secrets Manager

The **ssh-add** tool stores your SSH keys in AWS Secrets Manager with rich metadata:

```
# Basic usage - adds key to AWS Secrets Manager
ssh-add ~/.ssh/mykey_ed25519 [property-name] [region]

# Add both private and public keys
ssh-add ~/.ssh/mykey_ed25519 --pub

# Only upload the specified file, skip public key and fingerprint
ssh-add ~/.ssh/mykey_ed25519 --only

# Add key to both AWS Secrets Manager and your local SSH agent
ssh-add ~/.ssh/mykey_ed25519 --pub

# Skip adding to SSH agent
ssh-add ~/.ssh/mykey_ed25519 --no-ssh-add

# Enable verbose debug output
ssh-add ~/.ssh/mykey_ed25519 --debug
```

Key Metadata Features

Each key stored by **ssh-add** automatically includes metadata with:

- Key type (ed25519, RSA, etc.) and size (bits)
- Creation and update timestamps
- Version tracking for key rotation history
- SSH key fingerprint for agent identification
- Recommended rotation date (90 days from upload)
- Comments from the original key

Use SSH Keys with **ssh**

The **ssh** command retrieves keys from AWS Secrets Manager and uses them with SSH:

```
# Basic syntax
ssh user@hostname key-name [region] [options]
```

```
# Example: Connect to a server using a specific key
ssh ubuntu@my-server mykey_ed25519 us-east-2

# Enable debug output
ssh ubuntu@my-server mykey_ed25519 --debug
```

The **ssh** command performs the following steps:

1. Securely retrieves the key from AWS Secrets Manager
2. Identifies key fingerprint from metadata
3. Checks if the key is already loaded in **ssh-agent**
4. Adds the key to **ssh-agent** in memory (no disk writes) if needed
5. Connects to the specified server

Manage Keys with **ssh-admin**

The **ssh-admin** tool helps you manage your secrets in AWS Secrets Manager:

```
# Interactive mode
ssh-admin

# List all keys in the specified secret
ssh-admin --list
ssh-admin -l

# Specify AWS region
ssh-admin --region us-east-2
ssh-admin -r us-east-2

# Specify secret name
ssh-admin --secret my-ssh-keys
ssh-admin -s my-ssh-keys

# Show detailed information about a specific key
ssh-admin --detail mykey_ed25519
ssh-admin -d mykey_ed25519

# Enable debug output
ssh-admin --debug
```

Key Management Features

- Displays key metadata including type, size, and rotation schedule
- Shows key version history and update timestamps
- Views public key contents when available
- Creates new AWS Secrets Manager secrets if they don't exist
- Verifies appropriate AWS IAM permissions

Key Rotation Best Practices

Shh includes key rotation features:

- Each key automatically has a recommended rotation date (90 days after creation)
- The `ssh-admin --list` command shows when each key is due for rotation
- When a key is updated, the rotation history is preserved
- Version numbers are incremented automatically on each update

To rotate a key:

1. Generate a new SSH key: `ssh-keygen -t ed25519 -f ~/.ssh/new_key`
2. Add it to AWS Secrets Manager: `ssh-add ~/.ssh/new_key mykey_ed25519 --pub`
3. The previous key data is preserved in the rotation history

AWS Region Configuration

Region priority (from highest to lowest):

1. Command-line argument (e.g., `ssh-add ~/.ssh/mykey_ed25519 mykey us-east-2`)
2. `SSH_REGION` environment variable
3. `AWS_REGION` environment variable
4. `AWS_DEFAULT_REGION` environment variable
5. Default fallback (us-east-2)

Debug Options

All three commands support a `--debug` flag for troubleshooting:

```
ssh user@host keyname --debug
ssh-add ~/.ssh/mykey --debug
ssh-admin --debug
```

Security Considerations

- No SSH keys are ever written to disk during retrieval
- Keys are securely transmitted from AWS Secrets Manager to SSH agent in memory
- All AWS connections use your authenticated AWS CLI credentials
- Key fingerprints are stored to verify agent-loaded keys without requiring passphrase entry

Open Source & Contributions

We welcome contributions, improvements, and suggestions for enhancements.

```
git clone https://github.com/your-username/shh.git
```

Pull requests and issues are welcome!



License

Shh is released under the **MIT License**.



Logo Idea

A minimalist **SSH keyhole with sound waves**, representing **secrets & security** in a silent yet powerful way. 🗝️ 🎵

Let me know if you want tweaks or enhancements, babe! 🤔 🔥 🚀